

Trust3R: Unifying Feed-Forward Pointmap Prediction and Evidential Learning for Trust-Aware 3D Reconstruction

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Abstract

Geometric foundation models hold promise for unconstrained dense geometry prediction from uncalibrated images; however, in current feed-forward designs, their predicted confidence scores are heuristic, lack probabilistic interpretation, and often fail to indicate where and how much the predicted geometry can be trusted. To fill this gap, we present **Trust3R**, a trust-aware 3D reconstruction framework that pairs a lightweight gated residual mean refinement with evidential learning to predict pointmap evidence under a Normal-Inverse-Wishart prior and yield a closed-form multivariate Student-t predictive distribution. This design provides probabilistically grounded pointmap uncertainty estimates while adding moderate inference overhead. We evaluate on diverse indoor and outdoor benchmarks, and compare against MAST3R’s built-in confidence map as well as common uncertainty-aware baselines spanning single-pass heteroscedastic regression and sampling-based methods such as MC dropout and deep ensembles. Experimental results show that Trust3R consistently improves risk-coverage and sparsification, and generally improves geometric accuracy. Trust3R consistently improves uncertainty ranking across benchmarks (e.g. on ScanNet++: 25% lower AURC and 41% lower AUSE), enabling uncertainty-aware weighting for downstream alignment and fusion.

1. Introduction

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Recent advancements in feed-forward 3D reconstruction have significantly reshaped the way geometric reasoning is

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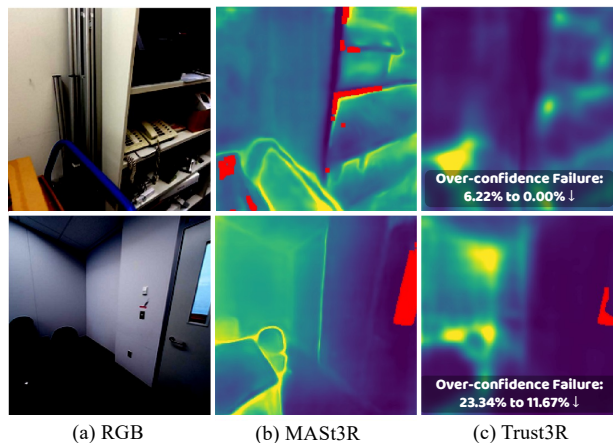


Figure 1. Reducing overconfident geometric failures with evidential uncertainty map: the 3D geometric pixels, which falls into the top- $q_{err}\%$ reconstruction errors while being assigned the lowest $q_{unc}\%$ uncertainty, are highlighted in red as overconfident failures. In contrast to the heuristic confidence in MAST3R, Trust3R reduces these incorrect yet highly confident regions, resulting in trust-aware uncertainty estimates for downstream weighting (e.g., alignment or fusion).

performed from images (Wang et al., 2024; Leroy et al., 2024). Instead of explicitly estimating camera parameters and enforcing multi-view consistency through iterative optimization, modern approaches directly regress dense 3D representations from image pairs or collections (Yao et al., 2018; Sun et al., 2021). DUST3R (Wang et al., 2024)/MASt3R (Leroy et al., 2024) established a particularly influential paradigm by grounding image matching directly in 3D: given a pair of images, the model predicts dense pointmaps whose geometric consistency implicitly encodes correspondence and depth.

However, feed-forward 3D reconstruction remains fundamentally ambiguous: occlusions, repetitive structures, low-texture regions, distribution shifts, and viewpoint degeneracies often admit multiple geometrically valid explanations (Wang et al., 2025a; Keetha et al., 2026; Wang et al., 2025b; Wang & Agapito, 2024). This ambiguity makes confidence assessment essential: beyond plausible geometry, we need predictions that indicate *where* they may fail. Accordingly, the focus is shifting from merely producing plausible 3D geometry to estimating its reliability (Laksh-

minarayanan et al., 2017; Kendall et al., 2018; Sattler et al., 2019; Besnier et al., 2021). Although MAST3R improves robustness compared to previous methods, it still predicts a single *deterministic* pointmap together with a per-pixel confidence score that acts as a learned reliability weight for the regression loss, rather than a predictive uncertainty with a clear probabilistic interpretation. As a result, confidence maps may not faithfully capture where and how much the geometry is reliable.

On the other hand, uncertainty quantification (UQ) offers a principled way to address this limitation (Blundell et al., 2015; Gal & Ghahramani, 2016; Lakshminarayanan et al., 2017; Kendall et al., 2018; Amini et al., 2020), but many widely used UQ paradigms rely on Bayesian sampling (Janssen, 2013) model inference, which becomes prohibitive for dense 3D pointmaps and conflicts with the real-time requirements of many real-world applications. Motivated by this, *evidence-based* uncertainty modeling offers a feed-forward alternative for uncertainty-aware 3D reconstruction: it directly predicts the parameters of a predictive distribution together with an explicit notion of evidence supporting the prediction, yielding closed-form uncertainty estimates with a clear probabilistic interpretation and no need for sampling, as in deep evidential learning (Sensoy et al., 2018). However, evidential learning is largely developed for low-dimensional regression with (approximately) independent outputs, and it is not plug-and-play for dense pointmap prediction. Vanilla evidential regression is designed for low-dimensional, independent multitasking regression outputs (Soleimany et al., 2021), whereas feed-forward pointmap reconstruction involves hundreds of thousands of correlated 3D points per image pair subject to strong geometric constraints and strict efficiency demands (Scharstein & Szeliski, 2002; Wang et al., 2024). Naively porting evidential heads to this setting can lead to unstable training and poorly estimated uncertainty with empirical geometric errors. This calls for a careful refinement strategy that preserves both the predictive accuracy and reliable uncertainty estimation.

To address the lack of reliable per-point uncertainty estimation in dense feed-forward pointmap prediction, we design **Trust3R**, a trust-aware 3D reconstruction framework based on evidential feed-forward pointmap prediction. First, Trust3R augments feed-forward multi-view geometry prediction with probabilistic outputs and an uncertainty estimate for each 3D point via a lightweight evidential UQ head, enabling the model to explicitly mark which regions of the 3D geometry are trustworthy and which remain uncertain, learned from data. Second, Trust3R introduces a gated refinement module that updates predictions through a residual connection, preserving the geometric accuracy and generalization of pretrained geometric foundation models while stabilizing evidential learning under challenging con-

ditions such as occlusions and distribution shifts. As shown in Figure 1, Trust3R produces efficient yet reliable uncertainty estimates that reduce overconfident geometric failures, making uncertainty directly usable for trust-aware filtering and weighting in downstream geometry. We summarize the following contributions in this paper:

- We introduce Trust3R, enabling geometric foundation models to produce probabilistically grounded uncertainty estimates for dense pointmaps via evidential learning. This supports explicit evaluation of geometric reliability under ambiguity and distribution shifts.
- Trust3R introduces pointmap-tailored designs for end-to-end evidential learning, combining an evidence-regularized predictive distribution with a gated residual refinement when needed. This preserves the strength of pretrained geometric foundation models while preventing overconfident uncertainty.
- We demonstrate the effectiveness of our predictive distribution and uncertainty estimates across a range of 3D perception benchmarks. Our method achieves a strong trade-off between model scalability and UQ quality while enabling practical reliability-aware filtering, alignment, and fusion.

2. Related Work

[Zihao: Meinert & Lavin 2021 should be cited here and discussed about it in following section]

3D geometric foundation models. Recent feed-forward 3D models regress geometry directly from images using large transformers, including depth- and correspondence-based approaches (Luo et al., 2016; Ranftl et al., 2021; Birkel et al., 2023) and pointmap formulations (Qian et al., 2022; Yu et al., 2022). DUST3R (Wang et al., 2024) predicts dense pairwise 3D pointmaps in a shared coordinate frame, while MAST3R (Leroy et al., 2024) improves robustness via mask-aware training and refined heads, achieving strong accuracy without iterative optimization. Pointmap representations offer an efficient and expressive backbone to downstream optimization modules, including Structure-from-Motion and visual SLAM systems (Schonberger & Frahm, 2016; Zhou et al., 2016; Dellaert & Kaess, 2017).

UQ in 3D geometry. Trust-aware 3D reconstruction has relied on heuristic confidence scores alongside dense pointmap predictions, but these scores lack a probabilistic interpretation and are often poorly estimated (Wang et al., 2024; Leroy et al., 2024; Kersting et al., 2007; Blundell et al., 2015). Bayesian approaches, including MC dropout (Gal & Ghahramani, 2016) and deep ensembles (Lakshminarayanan et al., 2017), offer uncertainty estimates by marginalizing model likelihood, yet they rely on multiple stochastic forward passes or multiple trained models (Carlin & Louis,

2008; Kruschke, 2010; Wang & Yeung, 2016). For dense 3D pointmap prediction with hundreds of thousands of correlated outputs per image pair, this sampling-based paradigm introduces prohibitive computational overhead. Conformal prediction (Shafer & Vovk, 2008; Angelopoulos et al., 2023) provides distribution-free uncertainty guarantees, but its worst-case nature often leads to overly conservative uncertainty regions and does not align with the fine-grained per-point 3D geometry (Mo et al., 2019; Fillioux et al., 2024). These limitations motivate evidential learning as a promising alternative (Sensoy et al., 2018; Amini et al., 2020; Gao et al., 2025): by predicting distributional parameters in a single forward pass, evidential methods offer probabilistically grounded uncertainty estimates with negligible inference overhead, making them particularly attractive for large-scale feed-forward 3D reconstruction.

3. Methodology

We consider the problem of dense 3D reconstruction from a set of images. Given one or more input images $\mathcal{I} = \{\mathcal{I}^v\}_{v=1}^N$, a 3D reconstruction model predicts a dense 3D point-cloud, denoted as *pointmap*, $\mathbf{X} = \{\mathbf{X}_i \in \mathbb{R}^3\}_{i=1}^M$, where each $\mathbf{X}_i$ represents a 3D point associated with a pixel or patch in the input images. In existing pointmap-based methods, such as DUS3R and MAST3R, $\mathbf{X}$ is treated as a deterministic output of a neural network. Figure 2 overviews our uncertainty-aware pipeline.

3.1. Preliminary

DUS3R and MAST3R formulate pairwise reconstruction as a feed-forward prediction problem. Given an image pair $(\mathcal{I}^1, \mathcal{I}^2)$, the network predicts two dense pointmaps (in a chosen reference frame) together with auxiliary maps used for matching and filtering

$$(\mathbf{X}^1, \mathbf{c}^1), (\mathbf{X}^2, \mathbf{c}^2) = f_{\text{pair}}(\mathcal{I}^1, \mathcal{I}^2), \quad (1)$$

where $\mathbf{X}^v \in \mathbb{R}^{H \times W \times 3}$ is the per-pixel pointmap prediction for view $v \in \{1, 2\}$, and $\mathbf{c}^v \in [0, 1]^{H \times W}$ is a learned confidence/validity score.

Importantly, the confidence in DUS3R/MAST3R is *not* a predictive uncertainty with a probabilistic interpretation (e.g., variance of a likelihood). Instead, it is a learned scalar that serves as a heuristic reliability indicator, typically used to (i) down-weight or filter unreliable pixels during training and/or (ii) select pixels for downstream alignment and fusion. As a consequence, such scores do not naturally support uncertainty propagation across views and can be poorly aligned with geometric errors under occlusions, textureless/reflective regions, and distribution shifts. This motivates replacing heuristic confidence with a predictive distribution whose uncertainty can be interpreted and consumed by downstream geometry modules.

3.2. Uncertainty-aware Pointmap Reconstruction

While MAST3R refines pointmap regression through geometrical and feature matching, the remaining valid pixels are still deterministic predictions but exhibit varying degrees of uncertainty due to noise, limited visual evidence, and inherent ambiguity in pairwise reconstruction.

For these reasons, we propose to substitute the deterministic pointmap 3D head with an *uncertainty-aware evidential head* that models each pointmap as a distribution rather than a single estimate. Specifically, for each pixel i and view v , the UQ head predicts a conditional distribution, $p(\mathbf{X}_i^v | \mathcal{I})$, capturing both the predicted pointmap and its associated uncertainty.

We assume that every 3D point $\mathbf{X}_i$ is drawn from a multivariate Gaussian likelihood

$$\mathbf{X}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i), \quad i \in \Omega, \quad (2)$$

where $\boldsymbol{\mu}_i \in \mathbb{R}^3$ is the mean of 3D points and $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3 \times 3}$ is a covariance matrix encoding spatial uncertainty, and Ω denotes the set of all 3D points. Rather than predicting $(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ directly, we adopt evidential learning in which the network predicts *distributional parameters* governing a family of predictive distributions.

3.3. Evidential UQ Head and Loss

To probabilistically estimate uncertainty over $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, we place a conjugate prior over these parameters. Assuming the multivariate Gaussian likelihood, this leads to a Normal–Inverse–Wishart (NIW) prior over the unknown mean and covariance:

$$\boldsymbol{\mu} | \boldsymbol{\Sigma} \sim \mathcal{N}(\mathbf{m}, \boldsymbol{\Sigma}/\kappa), \quad (3)$$

$$\boldsymbol{\Sigma} \sim \mathcal{W}^{-1}(\boldsymbol{\Psi}, \nu), \quad (4)$$

where $\mathbf{m} \in \mathbb{R}^3$ is the prior mean, $\kappa > 0$ controls the strength of belief in $\mathbf{m}$. $\mathcal{W}^{-1}(\cdot)$ is the inverse Wishart distribution, $\boldsymbol{\Psi} \in \mathbb{R}^{3 \times 3}$ is a positive-definite scale matrix, and $\nu > d + 1$ ($d = 3$) denotes the degrees of freedom. We denote the set of *evidential parameters* by

$$\boldsymbol{\theta} := \{\mathbf{m}, \kappa, \boldsymbol{\Psi}, \nu\}. \quad (5)$$

Under this formulation, the posterior distribution takes the form of the NIW distribution with joint density:

$$p(\boldsymbol{\mu}, \boldsymbol{\Sigma} | \boldsymbol{\theta}) = \mathcal{N}\left(\boldsymbol{\mu} | \mathbf{m}, \frac{\boldsymbol{\Sigma}}{\kappa}\right) \mathcal{W}^{-1}(\boldsymbol{\Sigma} | \boldsymbol{\Psi}, \nu). \quad (6)$$

From Bayesian probabilistic theory, the “model evidence” (or marginal likelihood) is obtained by integrating out the Gaussian likelihood parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, which yields a multivariate Student- t predictive distribution over the pointmap:

$$p(\mathbf{X} | \boldsymbol{\theta}) = \text{St}\left(\mathbf{X} | \mathbf{m}, \frac{\boldsymbol{\Psi}(\kappa + 1)}{\kappa(\nu - 2)}, \nu - 2\right). \quad (7)$$

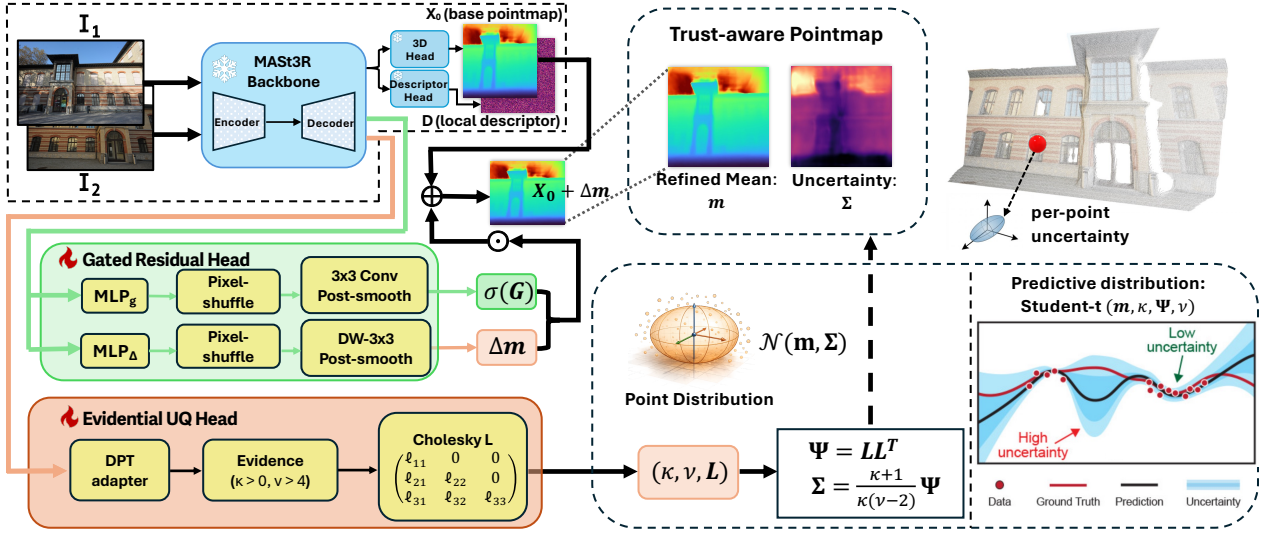


Figure 2. **Overview of the proposed Trust3R framework.** Built on a feed-forward MAST3R backbone, we augment pointmap prediction with a gated residual head for geometry refinement and an evidential UQ head that predicts the uncertainty-aware pointmap. This yields a closed-form multivariate Student- t predictive distribution, enabling per-point uncertainty estimation with negligible inference overhead. The evidential uncertainty can be directly exploited for trust-aware evaluation and uncertainty-weighted downstream alignment and fusion.

Evidential UQ head. Let $\mathbf{H}^v$ and $\mathbf{H}'^v$ denotes the embedded representations produced by the encoder and decoder of MAST3R’s ViT (Dosovitskiy et al., 2020), respectively. For each pixel $i \in \Omega$ and view $v \in \{1, 2\}$, the UQ head predicts the uncertainty parameters of a NIW distribution on the predicted point $\mathbf{X}_i^v$:

$$\{\kappa_i^v, \mathbf{L}_i^v, \nu_i^v\} = \text{Head}_{\text{UQ}}^v([\mathbf{H}^v, \mathbf{H}'^v]), \quad (8)$$

where $\mathbf{L}_i^v$ is a lower-triangular factor defined as the Cholesky decomposition of the positive-definite matrix Ψ_i^v . Instead of directly using the predictive mean $\mathbf{m}_i^v$ as output, we design a gated residual head to fine-tune the prediction from the pre-trained MAST3R backbone, which is deferred to Section 3.4 in detail.

The objective of the UQ head is to predict the distribution $p(\mathbf{X}^v | \mathcal{I}^v)$ of the whole pointmap. Following the mean-field assumption across predicted 3D points, we predict a variational distribution of independent 3D points

$$q(\mu^v, \Sigma^v | \theta^v) = \prod_{i \in \Omega} p(\mu_i^v, \Sigma_i^v | \theta_i^v), \quad (9)$$

parameterized by the UQ head in Eq. (8), without explicitly observing multiple samples per pixel.

Evidential UQ loss. Given the ground-truth 3D points $\hat{\mathbf{X}}_i^{v,1}$, training is performed by minimizing the negative log-likelihood (NLL) of the marginal predictive distribution in Eq. (7):

$$\mathcal{L}_{\text{NLL}} = - \sum_{v \in \{1,2\}} \sum_{j \in \Omega} \log p(\hat{\mathbf{X}}_j^{v,1} | \theta_j^v). \quad (10)$$

Minimizing the NLL alone admits degenerate solutions in which the model expresses unjustified certainty by inflating evidence parameters. To mitigate this problem, we introduce an evidence regularizer that penalizes excessive confidence when prediction errors are large.

Following the evidential learning framework, we define the total evidence for pixel i as $e_i = \kappa_i + \nu_i$, which jointly controls the concentration of the NIW distribution. The regularization term that penalizes high evidence under large prediction errors is defined as:

$$\mathcal{L}_{\text{evi}} = \sum_{v \in \{1,2\}} \sum_{i \in \Omega} \|\hat{\mathbf{X}}_i^{v,1} - \mathbf{m}_i^v\|^2 \cdot (\kappa_i^v + \nu_i^v). \quad (11)$$

The final evidential UQ loss for uncertainty-aware pointmap regression is given by

$$\mathcal{L}_{\text{UQ}} = \mathcal{L}_{\text{NLL}} + \lambda_{\text{evi}} \mathcal{L}_{\text{evi}}, \quad (12)$$

where λ_{evi} controls the strength of the evidence regularization. In practice, the overall training objective combines the base regression loss with $\mathcal{L}_{\text{UQ}}$ (via a convex mixing for stability). To avoid patch-grid artifacts introduced by token-to-pixel upsampling in the residual branch, we apply an identity-initialized post-upsampling smoothing module (Section A.3), and optionally regularize the spatial variation of the gating map with a weak total-variation prior.

3.4. Evidential Gate and Residual Head

The evidential NLL loss requires an accurate predictive mean to ensure stable optimization and avoid degenerate

uncertainty (e.g., trivially inflating variance to reduce NLL), which is crucial for reliable ranking under risk-coverage and sparsification. However, directly fine-tuning the mean prediction of a large pre-trained MAST3R model risks degrading its geometric accuracy and losing the benefits of pretraining. To preserve the original pointmap predictions while enabling uncertainty-aware learning, we seek a lightweight refinement mechanism that minimally perturbs the pretrained model.

Gated residual head. Let $\mathbf{X}_0^v \in \mathbb{R}^{H \times W \times 3}$ denote the pointmap predicted by the frozen MAST3R backbone for view v . Instead of modifying this prediction directly, we introduce a gated residual head that predicts a residual correction:

$$\Delta \mathbf{m}^v, \mathbf{G}^v = \text{Head}_{\text{GR}}^v([\mathbf{H}^v, \mathbf{H}^{v'}]), \quad (13)$$

along with a gating map $\sigma(\mathbf{G}^v) \in [0, 1]^{H \times W}$, which modulates the influence of the residual. The refined predictive mean is then given by

$$\mathbf{m}^v = \mathbf{X}_0^v + \sigma(\mathbf{G}^v) \odot \Delta \mathbf{m}^v, \quad (14)$$

where $\odot$ denotes element-wise multiplication broadcast over the 3D coordinates.

The gating mechanism allows the model to selectively adjust the pretrained mean only where necessary, while leaving confident and well-predicted regions unchanged. When $\sigma(\mathbf{G}^v) \approx 0$, the refined mean reduces to the original MAST3R prediction; when $\mathbf{G}^v$ has large positive logits (so $\sigma(\mathbf{G}^v) \approx 1$), the model is free to apply the predicted residual correction. This design ensures stable fine-tuning and prevents large deviations from the pretrained geometry.

To suppress patch-grid artifacts caused by token-to-pixel upsampling in the residual branch, we further apply a lightweight depthwise-separable post-smoothing operator after upsampling, initialized as an identity mapping so that the refinement starts from an exact no-op.

3.5. Predictive Mean and Uncertainty

In contrast to heuristic confidence maps, which assign only scalar scores that often lack a probabilistic interpretation, the evidential UQ head represents each pointmap as a full predictive distribution with well-defined uncertainty semantics. As a result, evidential uncertainty across views can be estimated analytically via Bayesian updates, rather than ad-hoc thresholding or reweighting.

The evidential formulation enables a principled decomposition of uncertainty: *aleatoric* and *epistemic*.

$$\underbrace{\mathbb{E}[\boldsymbol{\mu}] = \mathbf{m}}_{\text{prediction}}, \underbrace{\mathbb{E}[\boldsymbol{\Sigma}] = \frac{\Psi}{\nu - 4}}_{\text{aleatoric}}, \underbrace{\text{Var}[\boldsymbol{\mu}] = \frac{\Psi}{\kappa(\nu - 4)}}_{\text{epistemic}}. \quad (15)$$

Aleatoric uncertainty arises from observational noise and inherent ambiguity. Epistemic uncertainty is encoded through the evidence parameters (κ, ν) : low values correspond to weak evidence and higher model uncertainty, while large values indicate confident predictions supported by strong evidence. This distinction is particularly important in 3D reconstruction, where ambiguity may stem either from intrinsic scene properties (e.g., textureless regions) or from limited training coverage. This property allows pointmap uncertainty to be propagated and combined consistently in downstream inference modules, rather than being heuristically reweighted or thresholded.

Importantly, this formulation also yields uncertainty estimates in closed form and does not require Monte Carlo sampling at inference time. Each reconstructed point is associated with a compact set of evidential parameters, enabling efficient propagation of uncertainty through downstream geometric operations such as alignment, fusion, and filtering. This makes evidential pointmap reconstruction particularly suitable for large-scale and real-time 3D systems.

4. Experiments

In this section, we present the evaluation of Trust3R on indoor/outdoor benchmarks. We examine both uncertainty ranking quality and reconstruction accuracy, and report efficiency comparisons to quantify compute overhead. We further provide ablation studies to isolate the effects of key design choices, and include qualitative visualizations to illustrate typical failure modes. Additional experimental results including probabilistic scoring results (NLL) are deferred to the appendix.

4.1. Experimental Setup

For each image, we compute per-pixel 3D point errors and associate each pixel with a scalar uncertainty score. Unless noted otherwise, we apply a per-image Sim(3) alignment between predicted and ground-truth pointmaps before computing geometry-based errors, so MAE/RMSE and ranking-based UQ metrics (AURC (Geifman et al., 2018)/AUSE (Ilg et al., 2018)/Spearman ρ (Spearman, 1961)) reflect local geometric reliability rather than global pose/scale drift. All methods are evaluated on the same validity mask defined by available ground-truth depth/pointmap. We compute metrics per image and report dataset averages (NLL, when reported, is computed without alignment).

Datasets. We evaluate on diverse benchmarks chosen to stress-test uncertainty estimation across different capture conditions, spanning both indoor and outdoor scenes. Specifically, we use ScanNet++ (Yeshwanth et al., 2023) and TUM RGB-D (Sturm et al., 2012) for cluttered indoor RGB-D sequences with frequent occlusions and texture-

Table 1. Uncertainty evaluation on ScanNet++, TUM RGB-D, and KITTI, computed over valid pixels. We report AURC, AUSE, and Spearman ρ . Single-pass uses one inference pass, while multi-pass uses T stochastic inference passes or K trained models. Time reports training time in hours (h). Avg. reports the average measurement across the three datasets. **Bold** and underline mark the best and second-best results within the single-pass block.

Inference	Method	Time	Avg.			ScanNet++ (Indoor)			TUM RGB-D (Indoor)			KITTI (Outdoor)		
			AURC↓	AUSE↓	ρ ↑	AURC↓	AUSE↓	ρ ↑	AURC↓	AUSE↓	ρ ↑	AURC↓	AUSE↓	ρ ↑
Sampling	MCD ($T=16$)	–	0.4902	0.2726	0.2644	0.1777	0.0989	0.1937	0.0672	0.0368	0.2747	1.2257	0.6821	0.3249
	DeepEns ($K=5$)	100.2	0.2992	0.0916	0.4556	0.0722	0.0381	0.3921	0.0453	0.0208	0.3127	0.7802	0.2159	0.6620
Single-pass	MASt3R	–	0.4155	0.2053	0.4057	0.1649	0.0747	0.2837	0.0538	0.0233	0.4812	1.0277	0.5178	0.4523
	Hetero	9.4	<u>0.3973</u>	<u>0.1872</u>	<u>0.4183</u>	<u>0.1616</u>	<u>0.0715</u>	<u>0.3545</u>	0.0555	0.0251	0.4669	0.9749	<u>0.4650</u>	0.4335
	Trust3R (ours)	10.5	0.3861	0.1684	0.4898	0.1233	0.0444	0.4930	0.0481	0.0178	0.5169	<u>0.9868</u>	0.4431	0.4596

Table 2. Reconstruction accuracy measured by pointmap 3D point error. We report MAE and RMSE on ScanNet++ and TUM RGB-D, computed over valid pixels. **Avg.** is the mean over these benchmarks. **Bold** marks the best result in each column; smaller errors are better.

Method	Avg.		ScanNet++		TUM RGB-D	
	MAE↓	RMSE↓	MAE↓	RMSE↓	MAE↓	RMSE↓
MASt3R	0.1551	0.2313	0.2164	0.3026	0.0938	0.1600
Trust3R	0.1416	0.2173	0.1959	0.2849	0.0873	0.1496

Table 3. Computational overhead. We report training wall-clock time, peak GPU memory, and wall-clock inference time per pair. MASt3R checkpoint is pretrained, so training time is not comparable (–). For deep ensembles, training time is summed over K independently trained models.

Method	Training (h)	PeakMem (GB)	Inference (ms/pair)
MCD ($T=16$)	–	3.15	1225.77
DeepEns ($K=5$)	100.2	13.60	316.88
MASt3R	–	2.71	49.35
Hetero	9.4	2.88	70.14
Trust3R	10.5	3.16	80.92

poor regions, and KITTI (Geiger et al., 2013) and ETH3D (ablation mainly) (Schops et al., 2019) for outdoor/large-scale imagery with larger depth ranges, motion, and greater variability in scene structure and imaging conditions.

Compared methods. We compare against several single-pass baselines: **MASt3R** (Leroy et al., 2024), which uses the model’s predicted confidence score as a heuristic uncertainty signal; heteroscedastic Gaussian (Hetero, Lázar-Gredilla & Titsias, 2011), which predicts per-pixel variance in a single forward pass. We also compare against sampling-based references including MC Dropout (MCD, Gal & Ghahramani, 2016), which averages the prediction over T stochastic samples; Deep Ensembles (DeepEns, Lakshminarayanan et al., 2017), which aggregates K independently trained models. Sampling-based UQ methods typically provide stronger uncertainty estimates but incur substantially higher training and inference cost.

Metrics. We adopted three commonly used metrics for UQ evaluation: area under the risk-coverage curve (AURC),

area under the sparsification error curve (AUSE), and Spearman’s rank correlation coefficient (Spearman ρ). The definitions are provided in Appendix C. Reconstruction accuracy is also reported with MAE/RMSE over valid pixels using Sim(3)-aligned 3D errors.

Implementation details. To ensure a fair comparison, all uncertainty baselines share the same MASt3R backbone, training data, and optimization schedule unless otherwise noted. We freeze the backbone and train only lightweight add-on heads. Optimization uses AdamW (weight decay 0.05) (Loshchilov & Hutter, 2017) with base learning rate 3×10^{-4} (scaled by total batch size), with a cosine schedule over 10 epochs and no warmup. For evidential learning, we set evidence regularization $\lambda_{\text{evi}} = 10^{-3}$. Details can be found in Appendix A.5.

4.2. Uncertainty-aware Pointmap Prediction Results

Uncertainty evaluation. Table 1 reports uncertainty ranking performance across benchmarks. Figure 3 visualizes the corresponding risk-coverage and sparsification curves on three test sets, where Trust3R consistently yields lower ranking errors across a wide range of coverages. Among single-pass methods, Trust3R consistently achieves stronger selective ranking of high-error pixels, reflected by lower AURC and AUSE together with higher rank correlation. On ScanNet++, for instance, Trust3R significantly improves all three metrics, reducing AURC from 0.1649 to 0.1233 and AUSE from 0.0747 to 0.0444, while increasing rank correlation ρ from 0.2837 to 0.4946. Figure 4 further illustrates typical cases, showing that Trust3R’s predictive uncertainty aligns more closely with high-error regions than heuristic confidence scores in MASt3R.

Reconstruction accuracy. Apart from stronger uncertainty ranking quality in Table 1, Trust3R also improves MAE/RMSE of geometric accuracy on multiple datasets, as shown in Table 2. On both ScanNet++ and TUM RGB-D, the use of gated residual refinement leads to lower MAE and RMSE, while on ETH3D it yields consistent gains under domain shift. On KITTI, mean refinement can slightly increase MAE/RMSE relative to the frozen-mean baseline,

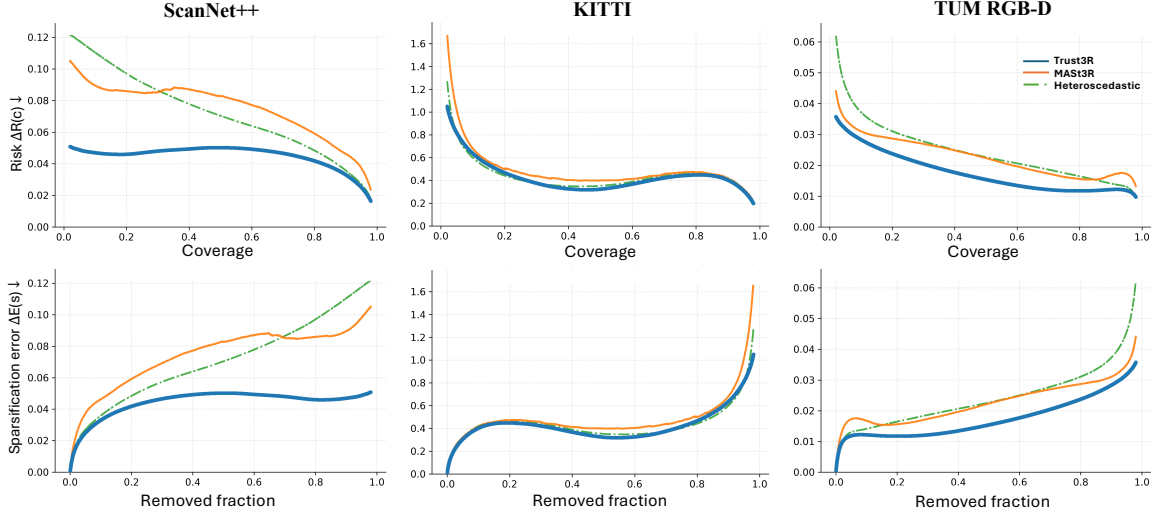


Figure 3. **Uncertainty ranking quality on three test sets.** Top row: Risk-Coverage $\Delta R(c) = R_{\text{unc}}(c) - R_{\text{oracle}}(c)$, where R_{oracle} is obtained by sorting pixels by their *true* 3D error for the same method (method-specific oracle). Bottom row: Sparsification error $\Delta E(s) = E_{\text{unc}}(s) - E_{\text{oracle}}(s)$. Lower is better, $\Delta = 0$ indicates perfect (oracle) ranking. Each subplot uses its own y-axis range for readability.

Table 4. **Uncertainty measure ablation on ETH3D.** We compare aleatoric u_{alea} , epistemic u_{epi} , and total u_{total} measures derived from the same predictive distribution.

Readout	AURC↓	AUSE↓	$\rho \uparrow$
u_{alea}	0.3175	0.1452	0.3093
u_{total}	<u>0.3064</u>	<u>0.1341</u>	<u>0.3455</u>
u_{epi}	0.3040	0.1318	0.3483

while uncertainty ranking remains improved, indicating that the uncertainty signal still prioritizes unreliable pixels even when the mean is imperfect.

Efficiency. Table 3 compares compute overhead. Trust3R remains single-pass (one model, one forward), matching the deployment cost of other single-pass baselines. In contrast, MC Dropout requires $T \times$ forwards and Deep Ensembles require $K \times$ models and forwards, making them substantially more expensive at inference.

Ensemble inference comparison. Deep Ensembles typically yield the strongest uncertainty metrics, but require training and running multiple models. Trust3R closes a substantial portion of this gap while remaining single-pass, which is important for downstream pipelines that cannot afford $K \times$ or $T \times$ inference. This makes Trust3R a practical drop-in uncertainty signal when downstream modules require aggressive pruning under a fixed compute budget.

4.3. Cross-Architecture Validation on VGGT

To evaluate the cross-architecture generalization of our method, we attach the same lightweight uncertainty head to a frozen VGGT backbone and keep the training proto-

Table 5. **Ablation on evidential variational family.** NIW models cross-axis correlations (full covariance), while NIG assumes independent coordinates (diagonal covariance).

Test set	Variant	AURC↓	AUSE↓	$\rho \uparrow$
ScanNet++	Trust3R (NIG)	0.1237	0.0448	0.4875
	Trust3R (NIW)	0.1233	0.0444	0.4930
ETH3D	Trust3R (NIG)	0.3213	0.1493	0.3229
	Trust3R (NIW)	0.3040	0.1318	0.3483

Table 6. **Generalization to VGGT.** Higher is better for ρ ; lower is better for AURC, AUSE, and NLL.

Variant	$\rho \uparrow$	AURC↓	AUSE↓	NLL↓
VGGT confidence	0.3162	0.1084	0.0452	-2.4770
VGGT + Trust3R	0.6419	0.0841	0.0209	-3.5999

col unchanged. We compare against the original VGGT confidence signal using the same uncertainty-ranking and probabilistic scoring metrics. As shown in Table 6, adding the proposed head improves both uncertainty ranking and NLL on VGGT, suggesting that the design is portable beyond a single MAST3R backbone.

4.4. Ablations

[Zihao: gated residual ablation across datasets should be added in this section with extra table and analysis] Our ablation study on ETH3D and ScanNet++ examines two design choices that impact the reliability of uncertainty ranking: (i) the uncertainty measure used for ranking, and (ii) the probabilistic head parameterization. Results are summarized in Table 4 and Table 5.



Figure 4. **Qualitative uncertainty comparison.** (a) RGB input; (b) GT 3D point error; (c) mismatch map for MASt3R confidence; (d) mismatch map for Trust3R predictive uncertainty. Mismatch is defined as $d(i) = |p_u(i) - p_e(i)|$, where p_u and p_e are percentile ranks of uncertainty and GT error over valid pixels (lower is better). Darker pixels indicate lower mismatch (better uncertainty–error alignment), meaning unreliable high-error regions are more likely to be assigned high uncertainty and thus removed by uncertainty-based filtering.

As shown in Table 4, we ablate the uncertainty measure used for ranking on ETH3D. We compare aleatoric uncertainty u_{alea} , total uncertainty u_{total} , and epistemic uncertainty u_{epi} , all computed from the same predictive distribution. Epistemic uncertainty provides the most reliable ranking signal: it attains the lowest AUC of 0.3040 and the highest Spearman correlation of 0.3483. We therefore use u_{epi} as the default ranking measure across benchmarks for consistency.

Table 5 further ablates the head design under the same ranking measure, comparing NIG with diagonal covariance and NIW with full covariance. NIW is consistently better on both benchmarks. For example, on ETH3D it reduces AUSE from 0.1493 to 0.1318, and on ScanNet++ it improves Spearman correlation from 0.4875 to 0.4930. This supports that modeling cross-dimension correlations improves the robustness of uncertainty ranking, especially when the test data deviates from training.

4.5. Downstream Evaluation: Reliability on Transparent Scenes

[Zihao: adding MASt3R-SLAM downstream experiment section/result here]

The reliability of 3D geometry affects many downstream

Table 7. **Transparent-scene reliability on Tricky24 (ring).** Mean over 32 scenes. Higher AUROC and lower FPR@95%TPR are better. Additional visualizations are provided in Appendix E.2.

Trust signal	AUROC↑	FPR@95%TPR↓
MASt3R	0.4523	0.9146
Trust3R	0.4786	0.8788

applications. Here, we evaluate on Tricky24, which contains transparent and reflective objects and scenes where, for example, non-Lambertian materials such as glass or metal can severely degrade depth sensing, a common challenge in robotic automation (Zama Ramirez et al., 2024). In this part, we use mask-based reliability metrics (AUROC and FPR@95%TPR) computed on a boundary ring band ($r=3$ pixels); details are in the appendix. We treat MASt3R confidence and our NIW uncertainty as anomaly scores and measure how well they highlight annotated transparent regions. As shown in Table 7, Trust3R achieves higher AUROC and lower FPR@95%TPR over 32 scenes, indicating better OOD awareness near refraction-induced boundaries.

5. Discussion and Limitations

Trust3R models per-point uncertainty with a unimodal predictive distribution, which cannot explicitly represent multi-hypothesis geometry in severely ambiguous regions. NIW prior of evidential uncertainty adds complexity to spatial correlations but may be less stable to optimize in challenging cases. Uncertainty may saturate in large textureless/occluded regions, trading coverage for selectivity under aggressive pruning.

6. Conclusion

We presented Trust3R, a trust-aware 3D reconstruction framework that equips dense feed-forward pointmap prediction with accurate and probabilistically grounded uncertainty estimates. By combining evidential learning with multi-view geometry prediction, Trust3R yields a closed-form Student- t predictive distribution whose uncertainty aligns closely with empirical geometric error, enabling reliable selective pointmaps. Across benchmarks, Trust3R consistently improves risk-coverage and sparsification behavior among single-pass methods without sacrificing geometric accuracy. Finally, our evidential uncertainty is used as an actionable reliability score for out-of-distribution awareness and failure-case identification, improving robustness in downstream reconstruction rather than merely producing qualitative uncertainty maps.

Impact Statement

[TODO: need revision.] This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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[Zihao: appendix todo: 1. organized appendix; 2. siuW three completeness results should be added to appendix; 3. mean prediction visualization; 4. real uncertainty heatmap depends on performance]

A. Additional Method Details

Given an input image pair (I_1, I_2) , the backbone predicts dense per-pixel 3D pointmaps $\hat{\mathbf{X}}_0^v \in \mathbb{R}^{H \times W \times 3}$ (view $v \in \{1, 2\}$) and a confidence map (when available). Our uncertainty heads output either (i) factorized evidential parameters (NIG) or (ii) multivariate evidential parameters (NIW). We denote a per-pixel 3D target as $\mathbf{x} \in \mathbb{R}^3$ (e.g., the aligned GT point in evaluation) and a scalar target as $y \in \mathbb{R}$ (e.g., metric depth).

Uncertainty readout. Many metrics require a scalar uncertainty score u_i per pixel. Unless noted otherwise, we use: (i) confidence $\rightarrow$ uncertainty via $u_{\text{conf}} = -\log(\text{conf} + \epsilon)$ (with small ϵ), (ii) XYZ variance $\rightarrow$ scalar via a reduction (default: trace) $u = \sum_{c \in \{x, y, z\}} \text{Var}[x_c]$. This matches our evaluation code setting `conf_unc_mode=neglog` and `xyz_reduce=trace`.

A.1. Multivariate Evidential Modeling via Normal-Inverse-Wishart (NIW)

To model cross-coordinate correlations, we use a Normal-Inverse-Wishart (NIW) evidential head per pixel:

$$\mathbf{m} \in \mathbb{R}^3, \quad \kappa > 0, \quad \nu > d + 1 \ (d = 3), \quad \Psi \in \mathbb{R}^{3 \times 3}, \quad \Psi \succ 0.$$

We enforce $\nu > d + 1$ in practice to ensure the predictive variance exists (equivalently $\nu_{\text{pred}} = \nu - (d - 1) > 2$).

Predictive distribution (Student- t). The induced predictive distribution is a multivariate Student- t :

$$\begin{aligned} p(\mathbf{x} \mid \mathbf{m}, \kappa, \nu, \Psi) &= \text{St}_d(\mathbf{x}; \mathbf{m}, \Sigma, \nu_{\text{pred}}), \\ \nu_{\text{pred}} &= \nu - (d - 1), \\ \Sigma &= \frac{\kappa + 1}{\kappa \nu_{\text{pred}}} \Psi. \end{aligned} \tag{16}$$

Here Σ is the *Student- t scale matrix*. The predictive covariance exists for $\nu_{\text{pred}} > 2$ and equals $\text{Cov}[\mathbf{x}] = \frac{\nu_{\text{pred}}}{\nu_{\text{pred}} - 2} \Sigma$.

Parameterization (PD constraint). We parameterize Ψ via a Cholesky factor $\mathbf{L}$: $\Psi = \mathbf{L}\mathbf{L}^\top$, and regress the 6 lower-triangular entries per pixel (diagonal entries mapped through `softplus` to ensure positivity).

Multivariate student- t NLL. Let $\delta = (\mathbf{x} - \mathbf{m})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{m})$. Then the log-probability is:

$$\begin{aligned} \log p(\mathbf{x}) &= \log \Gamma\left(\frac{\nu_{\text{pred}} + d}{2}\right) - \log \Gamma\left(\frac{\nu_{\text{pred}}}{2}\right) \\ &\quad - \frac{1}{2} \left(d \log(\nu_{\text{pred}} \pi) + \log |\Sigma| \right) \\ &\quad - \frac{\nu_{\text{pred}} + d}{2} \log \left(1 + \frac{\delta}{\nu_{\text{pred}}} \right), \end{aligned} \tag{17}$$

and we minimize $\mathcal{L}_{\text{NLL}} = -\log p(\mathbf{x})$ over valid pixels.

NIW evidential regularizer. Following evidential learning, we penalize being confident while wrong using an error-evidence coupling:

$$\mathcal{L}_{\text{evi}} = \|\mathbf{x} - \mathbf{m}\|^2 (\kappa + \nu), \quad \|\mathbf{v}\|_{\Psi^{-1}} = \sqrt{\mathbf{v}^\top \Psi^{-1} \mathbf{v}}.$$

Total NIW loss:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{NLL}} + \lambda_{\text{evi}} \mathcal{L}_{\text{evi}}.$$

(Aleatoric/epistemic decomposition. NIW admits a natural decomposition (for $\nu > d + 1$):

$$\underbrace{\Sigma_{\text{alea}}}_{\text{data noise}} = \frac{\Psi}{\nu - d - 1}, \quad \underbrace{\Sigma_{\text{epi}}}_{\text{mean uncertainty}} = \frac{\Psi}{\kappa(\nu - d - 1)}, \quad \Sigma_{\text{tot}} = \Sigma_{\text{alea}} + \Sigma_{\text{epi}} = \frac{\kappa + 1}{\kappa(\nu - d - 1)} \Psi.$$

For scalar readout we use $\text{tr}(\cdot)$ (trace) by default.

A.2. Residual-Gated Mean Refinement

To improve mean prediction while preserving the pretrained geometry, we add a residual-gated refinement:

$$\hat{\mathbf{X}}_{\text{ref}}^v = \hat{\mathbf{X}}_0^v + \sigma(\mathbf{g}^v) \odot \Delta^v, \quad (18)$$

where $\mathbf{g}^v$ (gate logits) and Δ^v (residual) are predicted from concatenated encoder/decoder tokens using lightweight MLPs, then reshaped to pixel space via token-to-pixel upsampling (pixel shuffle with factor equal to the ViT patch size). We initialize the residual branch to near-zero (final-layer weights/biases ≈ 0) so Eq. (18) starts as an identity mapping. In our implementation we optionally detach $\hat{\mathbf{X}}_0^v$ when training uncertainty heads to avoid destabilizing the backbone mean.

A.3. Post-upsampling Smoothing for Residual-Gated Refinement

Token-to-pixel upsampling may introduce patch-grid artifacts. We therefore optionally apply a lightweight post-upsampling smoothing:

$$\Delta \tilde{\mathbf{m}}^v = S_{\Delta}(\text{Up}(\Delta_{\text{tok}}^v)), \quad (19)$$

$$\tilde{\mathbf{g}}^v = S_g(\text{Up}(\mathbf{g}_{\text{tok}}^v)), \quad (20)$$

$$\hat{\mathbf{X}}_{\text{ref}}^v = \hat{\mathbf{X}}_0^v + \sigma(\tilde{\mathbf{g}}^v) \odot \Delta \tilde{\mathbf{m}}^v, \quad (21)$$

where S_{Δ}, S_g are depthwise-separable 3×3 convolutions (optionally followed by 1×1). We initialize S_{Δ}, S_g to be close to identity (e.g., delta-like kernels) so that refinement remains a no-op at initialization. Optionally, we regularize the spatial variation of the gating map with a weak TV prior:

$$\mathcal{L}_{\text{gate-tv}} = \sum_v \|\nabla \sigma(\tilde{\mathbf{g}}^v)\|_1.$$

A.4. Mean-field Evidential Modeling (NIG) for Scalar and Factorized Coordinates

For scalar regression we use Normal-Inverse-Gamma (NIG) evidence:

$$\gamma \in \mathbb{R}, \quad \nu > 0, \quad \alpha > 1, \quad \beta > 0,$$

which induces a Student- t :

$$p(y \mid \gamma, \nu, \alpha, \beta) = \text{St}\left(y; \mu = \gamma, \lambda = \frac{\beta(1+\nu)}{\nu\alpha}, \text{df} = 2\alpha\right).$$

Predictive mean/variance:

$$\mathbb{E}[y] = \gamma, \quad \text{Var}[y] = \frac{\beta(1+\nu)}{\nu(\alpha-1)}.$$

NLL (scalar). We optimize the induced Student- t NLL:

$$\begin{aligned} \mathcal{L}_{\text{NLL}} &= \frac{1}{2} \log\left(\frac{\pi}{\nu}\right) - \alpha \log(2\beta(1+\nu)) \\ &\quad + \left(\alpha + \frac{1}{2}\right) \log(\nu(y-\gamma)^2 + 2\beta(1+\nu)) \\ &\quad + \log \Gamma(\alpha) - \log \Gamma\left(\alpha + \frac{1}{2}\right). \end{aligned} \quad (22)$$

Evidential regularizer (scalar).

$$\mathcal{L}_{\text{evi}} = |y - \gamma| (2\nu + \alpha), \quad \mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{NLL}} + \lambda_{\text{evi}} \mathcal{L}_{\text{evi}}.$$

Factorized NIG (independent coordinates). For dense 3D points $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$, we assume conditional independence:

$$p(\mathbf{x} \mid \Theta) = \prod_{c \in \{x, y, z\}} p(x_c \mid \gamma_c, \nu_c, \alpha_c, \beta_c),$$

and minimize the mean of three scalar NIG losses:

$$\mathcal{L} = \frac{1}{3} \sum_{c \in \{x, y, z\}} \mathcal{L}_{\text{Total}}(x_c; \gamma_c, \nu_c, \alpha_c, \beta_c).$$

This yields a diagonal variance map (per-coordinate) and a scalar uncertainty via trace by default.

Aleatoric/epistemic decomposition. For NIG, one convenient decomposition is:

$$\sigma_{\text{alea}}^2 = \frac{\beta}{\alpha - 1}, \quad \sigma_{\text{epi}}^2 = \frac{\beta}{\nu(\alpha - 1)}, \quad \sigma_{\text{total}}^2 = \sigma_{\text{alea}}^2 + \sigma_{\text{epi}}^2 = \frac{\beta(1 + \nu)}{\nu(\alpha - 1)}.$$

For XYZ, apply per coordinate and reduce by trace (sum) if needed.

A.5. Implementation Details

Backbone and heads. We build on an asymmetric MAST3R backbone and attach: (i) a multivariate NIW evidential head that predicts $(\kappa, \nu, \mathbf{L})$ per pixel; (ii) the GR mean-refinement head (with PixelShuffle upsampling and post-upsampling smoothing enabled).

NIW head parameterization. For each pixel, the head predicts raw outputs mapped as:

$$\kappa = \text{softplus}(\cdot) + \epsilon, \quad \nu = (d + 1) + \text{softplus}(\cdot) \quad (d=3),$$

and $\Psi = \mathbf{L}\mathbf{L}^\top$ where the diagonal entries of $\mathbf{L}$ use $\text{softplus}(\cdot) + \epsilon$ and off-diagonal entries are unconstrained. For uncertainty readout we use the diagonal of $\Sigma = \text{scale} \cdot \Psi$ (Sec. A.1), and reduce it by trace.

Training setup. We freeze the MAST3R backbone and geometry heads and train only the NIW UQ heads (with GR post-smoothing enabled). Optimization uses AdamW with weight decay 0.05 and base learning rate 3×10^{-4} , scaled by total batch size (effective batch size 10), with a cosine schedule over 10 epochs and no warmup. For evidential learning, we set $\lambda_{\text{uq}}^{xyz} = 0.05$ and evidence regularization $\lambda_{\text{evi}}^{xyz} = 10^{-3}$; all other UQ losses are disabled. Training uses the mix4 dataset (25k ScanNet++, 25k ARKitScenes, 50k Waymo, 50k MegaDepth; resolution 224 with aug_crop=16) and validates on 2k ScanNet++ (aug_crop=0).

Objective. Since the backbone/geometry are frozen, we optimize only the NIW uncertainty objective on valid pixels:

$$\mathcal{L} = \lambda_{\text{uq}} \cdot \left(\mathcal{L}_{\text{NLL}} + \lambda_{\text{evi}} \mathcal{L}_{\text{evi}} \right),$$

summed over both views and averaged over valid pixels.

B. Datasets and Preprocessing

We evaluate on both in-domain (ID) and out-of-domain (OOD) datasets. We form image pairs using dataset-provided pair lists or fixed-stride temporal pairing (see below), and keep train/val/test disjoint by scenes.

ScanNet++. We use the ScanNet++ preprocessed metadata that contains (scene, image, depth, intrinsics, pose) and a precomputed pair list. For training/validation, we split by scenes (scene-disjoint) following our loader: the first 90% scenes (sorted) are used for training and the remaining 10% for validation. RGB images and depth are resized to the network resolution; intrinsics are adjusted accordingly. We treat pixels with invalid depth (zero / missing) as invalid in training and evaluation masks.

HyperSim / TUM RGB-D / KITTI / ETH3D. We follow the standard evaluation protocol in the main paper. For each dataset, we form image pairs using the dataset-provided pair lists or fixed-stride temporal pairing, and apply the dataset validity mask (pixels without valid depth/points are excluded).

C. Evaluation Protocol and Metrics

We evaluate dense pointmap prediction with dataset-level aggregation over valid pixels. Pointmap errors are computed after the evaluation alignment step (e.g., Sim(3) in the main paper), while probabilistic scoring (NLL) is computed in the raw camera frame (without alignment).

Notation. For one image pair, the model predicts a dense pointmap $\hat{\mathbf{X}} \in \mathbb{R}^{H \times W \times 3}$ and we have ground truth $\mathbf{X}^* \in \mathbb{R}^{H \times W \times 3}$ in the same coordinate frame after the evaluation alignment step (e.g., Sim(3)). Let Ω denote valid pixels and $N = |\Omega|$. For each $i \in \Omega$, define the 3D error

$$e_i = \|\hat{\mathbf{x}}_i - \mathbf{x}_i^*\|_2, \quad (23)$$

and a scalar uncertainty score $u_i \geq 0$ (larger = more uncertain).

Uncertainty readout.

- **Confidence baseline:** $u_i = -\log(\text{conf}_i + \epsilon)$.
- **Ours (NIG/NIW):** use predicted per-coordinate variance (or NIW diagonal of the scale/covariance) and reduce by $u_i = \sum_{c \in \{x, y, z\}} \text{Var}[x_{i,c}]$ (trace).
- **MC Dropout / Deep Ensemble:** moment-match a diagonal Gaussian from sampled predictions and reduce by trace for u_i .

Pointmap accuracy.

$$\text{MAE}_{3D} = \frac{1}{N} \sum_{i \in \Omega} e_i, \quad (24)$$

$$\text{RMSE}_{3D} = \sqrt{\frac{1}{N} \sum_{i \in \Omega} e_i^2}. \quad (25)$$

Ranking correlation (Spearman’s ρ).

$$\rho = \text{corr}(\text{rank}(u), \text{rank}(e)), \quad (26)$$

computed over valid pixels (ties are rare in practice).

Risk-coverage curve and AURC. Retain the $\lfloor cN \rfloor$ pixels with smallest u_i ; risk is

$$R(c) = \frac{1}{|\Omega_c|} \sum_{i \in \Omega_c} e_i. \quad (27)$$

AURC is the area under $R(c)$ over $c \in (0, 1]$ (numerically integrated on a uniform grid; trapezoidal rule in our code).

Oracle risk-coverage. Oracle ranks by e_i ascending and gives a lower bound for AURC.

Sparsification curve and AUSE. Remove the top $\lceil sN \rceil$ pixels by uncertainty (largest u_i). Let $E_{\text{unc}}(s)$ be mean remaining error; oracle removes by e_i . AUSE is the area of the gap:

$$\text{AUSE} = \int_0^1 \left(E_{\text{unc}}(s) - E_{\text{oracle}}(s) \right) ds,$$

computed numerically (trapezoidal rule).

Negative log-likelihood (NLL). When a method defines a predictive distribution, we report mean NLL in the *raw camera frame*:

$$\text{NLL} = -\frac{1}{N} \sum_{i \in \Omega} \log p_{\theta}(\mathbf{x}_i^*). \quad (28)$$

Gaussian predictive models. For heteroscedastic regression: $p_{\theta}(\mathbf{x}_i) = \mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\boldsymbol{\sigma}_i^2))$. For MC Dropout / Deep Ensembles we moment-match a diagonal Gaussian and add σ_0^2 :

$$\boldsymbol{\sigma}_{i,\text{tot}}^2 = \boldsymbol{\sigma}_{i,\text{epi}}^2 + \sigma_0^2.$$

Table 8. Ablation: evidential head choice. We compare factorized XYZ-NIG and multivariate XYZ-NIW (same uncertainty readout: total/trace). Lower is better for AURC/AUSE; higher is better for Spearman ρ . (ScanNet++ in Table 1 setting.)

Variant	AURC↓	AUSE↓	ρ ↑
XYZ-NIG	0.1229	0.0441	0.4946
XYZ-NIW	0.1253	0.0464	0.4725

Table 9. Ablation: uncertainty sources (ETH3D). We compare total vs. decomposed uncertainty readouts (trace reduction) for XYZ-NIW.

Source	AURC↓	AUSE↓	ρ ↑
Total (trace)	0.3064	0.1341	0.3455
Aleatoric only	0.3175	0.1452	0.3093
Epistemic only	0.3040	0.1318	0.3483

Per-pixel NLL:

$$-\log p_{\theta}(\mathbf{x}_i^*) = \frac{1}{2} \sum_{d=1}^3 \left[\log(2\pi\sigma_{i,d}^2) + \frac{(x_{i,d}^* - \mu_{i,d})^2}{\sigma_{i,d}^2} \right]. \quad (29)$$

Evidential models. For NIG/NIW, we evaluate NLL under the induced predictive Student- t distribution (Sec. A.4, Sec. A.1). Confidence-only baselines do not define an explicit likelihood and are omitted from NLL.

D. Downstream Evaluation Details

D.1. Tricky24: Reliability on Transparent Scenes

We evaluate on 32 Tricky24 scenes with provided transparent-object masks, using per-view masks aligned to the evaluated view. We follow the uncertainty readout in Sec. C (conf $\rightarrow u$ via $-\log(\text{conf} + \epsilon)$; NIW via $u = \text{tr}(\Sigma_{\mathbf{x}}) / \text{trace reduction}$).

Boundary ring construction. To focus on boundary failures, we construct a ring band of radius $r=3$ pixels via morphology:

$$\text{ring} = \text{dilate}(m, r) \wedge \neg \text{erode}(m, r).$$

Metrics. For each scene, pixels in the ring are treated as positives and all remaining pixels as negatives. We compute AUROC and FPR95 (false positive rate at 95% true positive rate), and report the mean of per-scene scores in Table 7.

E. Additional Results

E.1. Ablations and Additional Metrics

Evidential head choice. Table 8 compares two evidential parameterizations under the same evaluation protocol as Table 1 (ScanNet++). We use the same uncertainty readout (total/trace) for a like-for-like comparison.

Uncertainty sources. Table 9 evaluates different uncertainty readouts for XYZ-NIW on ETH3D, including total (trace) and decomposed aleatoric/epistemic components.

Negative log-likelihood. Table 10 reports mean NLL across datasets for probabilistic baselines. For MC Dropout and Deep Ensemble, we moment-match a diagonal Gaussian and add a small σ_0^2 (selected on ScanNet++ validation). Confidence-only baselines do not define a likelihood and are omitted.

Table 10. NLL across datasets. We report mean negative log-likelihood (NLL) (lower is better). For MC Dropout and Deep Ensemble, we moment-match a diagonal Gaussian and add σ_0^2 (selected on ScanNet++ val). Confidence-only baselines do not define a likelihood and are omitted.

Method	ScanNet++	TUM	KITTI	ETH3D
Hetero (Gauss)	-0.8374	-4.2523	3.5102	37.4365
XYZ-NIG (ours)	-0.6852	-1.4672	1.1738	7.8879
XYZ-NIW (ours)	-3.0020	-5.1541	9.4236	11.6632

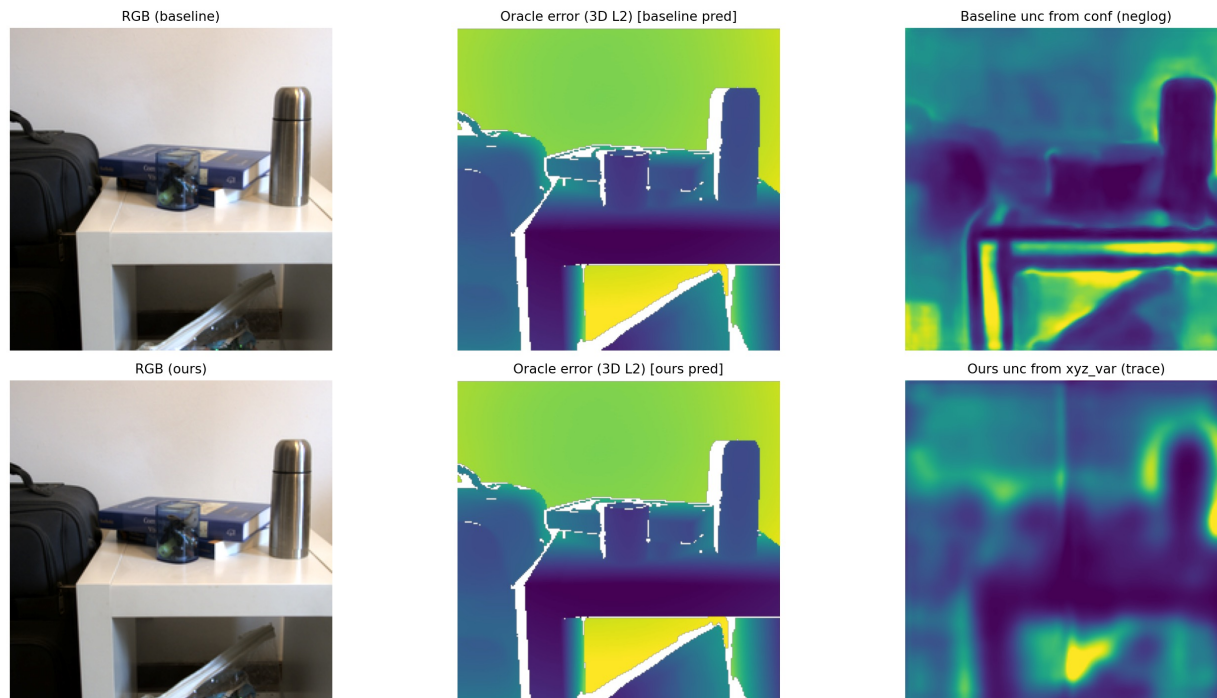


Figure 5. Qualitative reliability on Tricky24 (example A). Top: MAST3R baseline; bottom: Trust3R. Columns show the input RGB, oracle 3D error (L2), and the corresponding uncertainty map.

E.2. Qualitative Results

Transparent-scene reliability (Tricky24). Figures 5–6 provide additional examples on Tricky24. Each figure uses the same layout (top: MAST3R baseline; bottom: Trust3R) with columns showing RGB, oracle 3D error (L2), and the corresponding uncertainty map.

Mismatch-map visualization. Figures 7–8 visualize mismatch between uncertainty ranking and oracle error ranking (darker is better alignment).

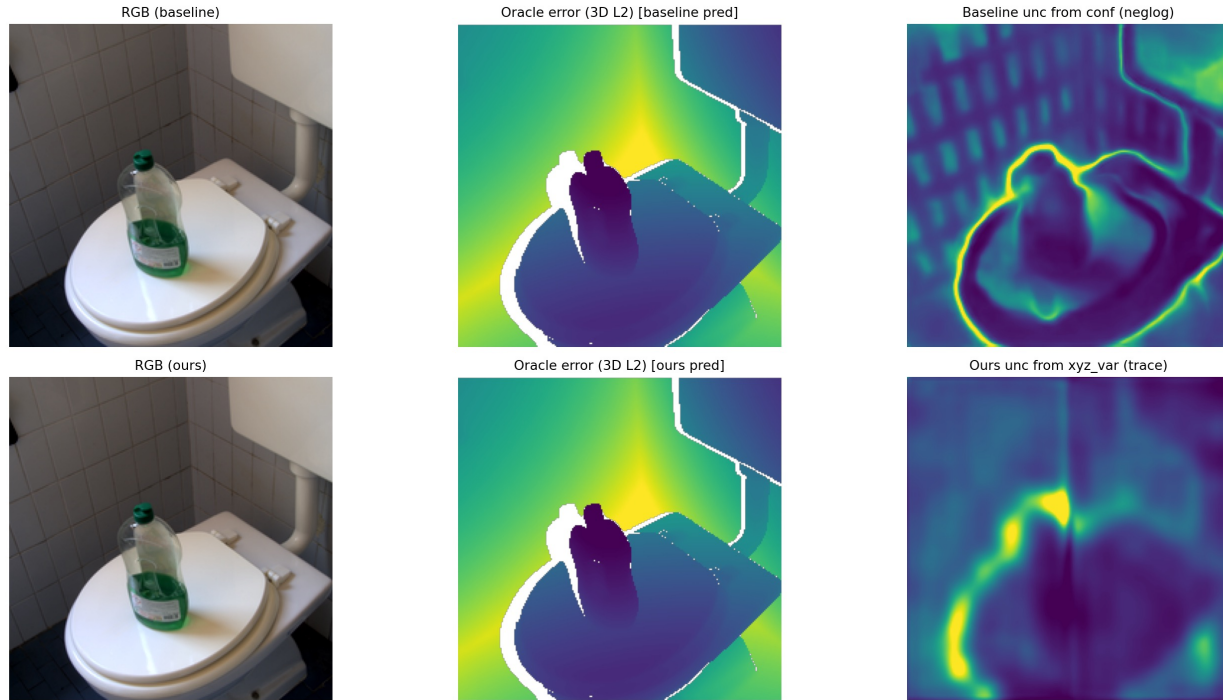


Figure 6. **Qualitative reliability on Tricky24 (example B).** Top: MAST3R baseline; bottom: Trust3R. Columns show the input RGB, oracle 3D error (L2), and the corresponding uncertainty map.

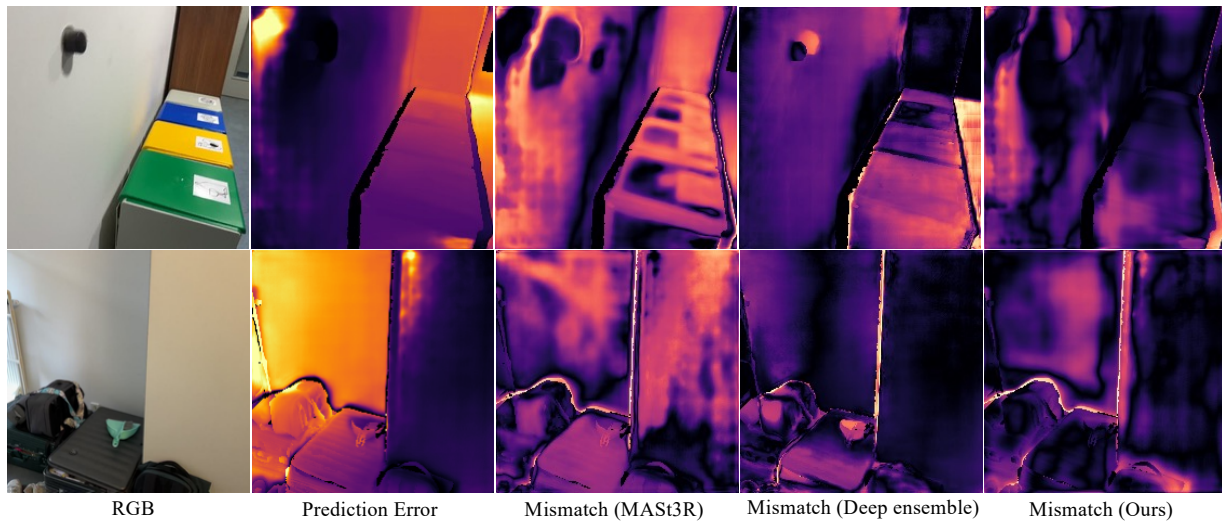


Figure 7. **Additional qualitative uncertainty comparison.** (a) RGB input; (b) GT 3D point error; (c) mismatch map for MAST3R confidence; (d) mismatch map for Trust3R predictive uncertainty.

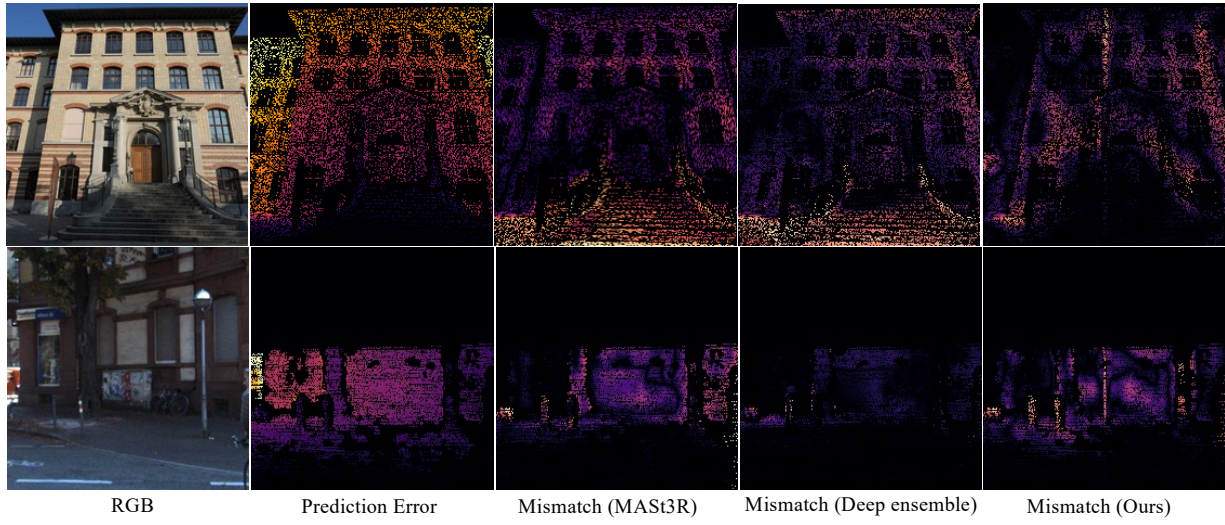


Figure 8. **Additional qualitative results.** More examples showing that Trust3R uncertainty aligns better with GT error compared to confidence, particularly on reflective/transparent regions and thin structures.

F. Reproducibility

- **Data splits.** We keep train/val/test disjoint by scenes and use fixed pairing strategies per dataset (Sec. B).
- **Metrics.** We report pointmap accuracy after alignment and compute NLL in the raw camera frame (Sec. C).
- **Hyperparameters.** We list optimization settings and the NLL noise floor selection σ_0^2 in Sec. A.5.